

Rubik's Cube Solver

Performance Statistics

1 Introduction

A set of 19 execution runs of the hybrid Rubik's Cube solver (Two-Phase G1 reduction + Korf IDA* completion) are tracked and across all mixed difficulty scrambles, the solver maintains an average runtime of **~1.27 seconds**. Additionally, the interactive simulator window requires an average of **7.67 seconds** to render the cube solving sequence.

2 Data Summary and Metrics

Table: Execution Metrics

Trial	Scramble Depth	Phase 1 Moves	Solution Length	Time (ms)
1	15	9	15	48
2	7	7	7	1
3	7	2	8	24
4	7	7	13	14
5	7	6	6	10
6	7	12	12	1,388
7	15	8	8	9
8	7	7	7	1
9	7	6	6	3
10	7	6	12	43
11	7	10	10	1,222
12	7	2	2	5
13	7	8	14	31
14	15	12	12	5,218
15	15	12	20	9,124
16	15	12	12	3,172
17	15	11	11	231
18	15	11	19	3,379
19	15	11	11	118

Overall Average Time Taken: 1265.32 ms